咖啡冲煮技能

实操考试（讲师版）

Brewing Professional

Practical Exam (Trainer)

一位专业冲煮师最强大的工具，莫过于他们善于分析的思维，以及处理各式冲煮变因的能力。这些冲煮变因有助于了解如何以最佳方式解读信息，然后提出解决方案或意见，以提高客户的咖啡质量、服务与出杯。

The most powerful tools that a Professional Brewer possesses are their analytical minds and their ability to process a multitude of changing variables. These variables help to understand how best to interpret the information and then to offer a solution or opinion that will improve the coffee quality, service and delivery for their clients.

学员必须先完成本考试的第 1 大题和第 2 大题，最后完成第 3 大题。

The learner must complete Sections 1 and 2 of this exam prior to Section 3.

- 在本次实操考试中，学员必须使用一台**重力式滴滤设备**冲煮所有咖啡。建议使用自动滴滤咖啡机冲煮，但也可以使用其它类型的重力式滴滤设备。
  A **drip filter brewer** must be used for all brews prepared in this practical exam. An automatic drip filter brewer is recommended, but any drip filter brewer is allowed.

- 本次实操考试中使用的咖啡熟豆，必须能够产生 24%的萃取率，并为课程活动中使用过的豆子，但一定没有在水质活动上使用过（3.03.07.08）。
  The roasted coffee used in this practical exam must be capable of producing 24% solubles yield and must have been used for some activities during the course. However, it must not have been used in the water activity (3.03.07.08).

- 学员每次提交咖啡成品时，**必须**填写并提交【冲煮参数和参数修正工作表】。
  The learner **must** complete and submit Brewing Professional Recipe Modification Worksheet for each brew submitted.

- 学员应于第 1 大题中，利用参考用咖啡与其它咖啡进行感官比较。这将有助于学员决定他们在第 3 大题的冲煮计划。
  The learner should use the reference brew for sensory comparison with other brews in section 1. This will help the learner to determine their brew plan in section 3.

实操考试整体学习成果 **Practical Exam Overall Learning Outcomes**		
第 1 大题 **Section 1**	**第 2 大题** **Section 2**	**第 3 大题** **Section 3**
• 能够**独立控制**浓度和萃取率，**熟练运用**冲煮控制图。 Competency in **navigating** the brewing control chart by **controlling** solubles concentration and solubles yield **independent** of each other.	• 能够**分析**咖啡冲煮用水在化学层面的关键方向。 Competency to **analyse** the key aspects of water chemistry for coffee brewing	• 能够**调整**冲煮计划，并**选择**一款能提升冲煮质量的水。 Competency to **adapt** a brew plan and **choose** a water that will improve the quality of brewed coffee

共三大题 **SECTION**	考试时间（分钟） **MAXIMUM TIME (MINS)**
1. 第 1 大题 - 浓度和萃取率比较（A、B 和 C 部分） **Strength and Extraction Comparison (Parts A, B, and C)**	
A 部分：参考用咖啡 **Part A: Reference Brew**	**15**
B 部分：咖啡萃取率比较 **Part B: Solubles Yield Comparison**	**60**
C部分：咖啡浓度比较 **Part C: Solubles Concentration Comparison**	**60**
2. 第 2 大题 – 冲煮用水分析 **Section 2 - Water Analysis**	
A 部分：冲煮用水分析 **Part A: Water Analysis**	**10**
3. 第 3 大题 - 提升冲煮咖啡质量（A、B 和 C 部分） **Section 3 - Improve Brew Quality (Parts A, B, and C)**	**45**
A 部分：分析和建议 **Part A: Analysis & Proposal**	
B 部分：咖啡质量预测 **Part B: Brew Prediction**	
C 部分：冲煮咖啡成果 **Part C: Brew Outcome**	
总计 **TOTAL**	**190 分钟** **190 MINS**

第 1 大题：浓度和萃取比较（A、B 和 C 部分）

Section 1: Strength and extraction comparison (parts A, B, and C)

总分 31 分| 总时间 135 分钟　31 total points possible | 135 total minutes

本大题将测验学员是否充分掌握和理解所有冲煮参数以及对于萃取的预测影响，从而熟练运用咖啡冲煮控制图。

This section of the practical exam will test the learner's ability to navigate the Coffee Brewing Control Chart by using their mastery and knowledge of all brewing parameters and their predicted effect on extraction.

- 目标咖啡（即 B1.B2.C1.C2）的条件设定落在『理想』范围之外，学员必须充分了解所需的控制参数/变因，以製作出符合设定条件的目标咖啡。The target brews are set outside the "Ideal" range, and learners must have a thorough understanding of the control measures required to produce the target brews.
- 学员必须先制作参考用咖啡（A 部分），方可继续 B 部分和 C 部分。

  Learners must first produce a reference brew (Part A) before progressing to Parts B and C.
- 接着要求学员冲煮 2 杯浓度相同、但萃取率不同的咖啡（B 部分），以及冲煮 2 杯萃取率相同、但浓度不同的咖啡（C 部分）。

  Learners are then required to prepare 2 brews with the same solubles concentration, but with different solubles yield (Part B), and also produce 2 brews with the same solubles yield, but with different solubles concentration (Part C).
- B 部分和 C 部分不必按顺序进行或于同一天完成，由 AST 酌情决定。

  Learners may attempt Parts B and C in any order and, at the discretion of the AST, on different days of the course.
- 在 B 部分和 C 部分中，学员需要解释自己制作每种目标咖啡的方式，包括对冲煮参数和/或技术上做出的更改。

  In Parts B and C, learners are required to explain how they produced each of the target brews, including the changes they made to brewing parameters and/or their technique.
- 学员应在【冲煮参数和参数修正工作表】中记录冲煮参数，并于下方记录合格案例。

  Learners should use Brewing Professional Recipe Modification Worksheet to record brew parameters, then record successful brews below.
- **注意**：学员只有在冲煮的咖啡达标时方需解释，不合格的咖啡及相关解释一律不予计分。

  Note: Learners may only provide an explanation for a brew they produce <u>if</u> that brew meets its target. Unsuccessful brews and their explanations will receive zero points.

第 1 大题 A 部分 参考用咖啡 Part 1.A Reference brew

考试时间：15 分钟 Time available: 15 minutes

- 学员在本次考试中必须使用重力式滴滤设备，并且使用同一设备冲煮所有咖啡。
  Learners must use a drip filter brewer for this brew, and the same device must be used for all brews prepared in this exam.
- 建议使用自动滴滤咖啡机。
  An automatic filter brewer is recommended.
- 参考用咖啡必须符合浓度 1.30%（+/-0.05%）、萃取率为 20%（+/-1.00%）的条件。
  The reference brew must have a solubles concentration of 1.30% (+/- 0.05%), with a solubles yield of 20% (+/- 1.00%).
- 学员必须描述咖啡的香气、味道和醇厚度。
  The learner must provide descriptions for aroma, taste, and body aspects of the brew.

	研磨咖啡粉量 Ground Coffee Dose (g/oz)	用水量 Water Dose (g/oz)	冲煮时间 Brew Time	浓度 Solubles Concentration %	萃取率 Solubles Yield %	成功冲煮咖啡所需的时间：____:____ Time taken to prepare successful brew: ____:____	分数 Points
参考用咖啡 A Reference Brew - A			:			合格咖啡可得 2 分 2 points for successful brew	
咖啡描述 Brew Description							
香气（风味）： Aroma (Flavour):						描述得宜可得 1 分 1 point for suitable descriptor	
味道（酸度）： Taste (Acidity):						描述得宜可得 1 分 1 point for suitable descriptor	
味道（甜度）： Taste (Sweetness):						描述得宜可得 1 分 1 point for suitable descriptor	
味道（苦味）： Taste (Bitterness):						描述得宜可得 1 分 1 point for suitable descriptor	
醇厚度： Body:						描述得宜可得 1 分 1 point for suitable descriptor	
					总分 = 7 Total points available = 7	总得分 Total points awarded	

第 1 大题 B 部分 咖啡萃取率比较
Part 1.B Solubles yield comparison

考试时间：60 分钟 Time available: 60 minutes

- 学员需冲煮“符合”下列标准的咖啡。
  The learner will create brews that “hit” the targets listed below.
- 学员必须解释如何及为何调整参考用咖啡中使用的参数，以达到咖啡 B1 和 B2 的要求。
  The learner must provide explanations for how and why they adjusted the parameters used in the reference brew to achieve Brews B1 and B2.

一致性的咖啡浓度：1.30%（+/- 0.05%） Uniform Solubles Concentration: 1.30% (+/- 0.05%)

- 咖啡 B1 - 16% 萃取率 (+/- 1.00%）
  Brew B1 - 16% Solubles Yield (+/- 1.00%)
- 咖啡 B2 - 24% 萃取率 (+/- 1.00%）
  Brew B2 - 24% Solubles Yield (+/- 1.00%)

咖啡 Brew	研磨咖啡粉量 Ground Coffee Dose (g/oz)	用水量 Water Dose (g/oz)	冲煮时间 Brew Time	浓度 Solubles Concentration %	萃取率 Solubles Yield %	成功冲煮咖啡所需的时间：____:____ Time taken to prepare successful brew: ____:____	分数 Points
B 1			:			合格咖啡可得 2 分 2 points for successful brew	

描述与参考用咖啡相比的感官差异： Describe sensory differences compared with reference brew:	
描述得宜可得 1 分 1 point for suitable description	

粉水比 **Coffee to Water Ratio**	您是否更改粉水比？**是 / 否** Did you change the coffee to water ratio? **Yes/No**	
为何更改 **或** 保留粉水比？ Why did you change the coffee to water ratio, **or** why did you leave the coffee to water ratio unchanged?		
解释合理可得 1 分 1 point for suitable explanation		

咖啡粉量及用水量 **Ground Coffee & Water Doses**	您是否更改咖啡粉量和/或用水量？**是 / 否** Did you change the ground coffee dose and/or water dose? **Yes/No**	
为何改变咖啡粉量和/或用水量？**或是** 为何不改变咖啡粉量和/或用水量？ Why did you change the ground coffee and/or water dose, **or** why did you leave the ground coffee dose and/or water dose unchanged?		
解释合理可得 1 分 1 point for suitable explanation		

其它参数/变量 **Other Control Measures**	您是否更改任何其它参数/变量？**是 / 否** Did you change any other control measures? **Yes/No**	
为何保留所有参数/变量不变？ **或是** 更改了哪些参数/变量及其原因？ Why did you leave all other control measures unchanged, **or** which control measures did you change, and why did you change them?		
解释合理可得 1 分 1 point for suitable explanation		
总分 = 6 Total points available = 6	总得分 Total points awarded	

咖啡 Brew	研磨咖啡粉量 Ground Coffee Dose (g/oz)	用水量 Water Dose (g/oz)	冲煮时间 Brew Time	浓度 Solubles Concentration %	萃取率 Solubles Yield %	成功冲煮咖啡所需的时间：____:____ Time taken to prepare successful brew: ____:____	分数 Points
B 2			:			合格咖啡可得 2 分 2 points for successful brew	

描述与参考用咖啡相比的感官差异：

Describe sensory differences compared with reference brew:

描述得宜可得 1 分

1 point for suitable description

粉水比 **Coffee to Water Ratio**	您是否更改粉水比？**是 / 否** Did you change the coffee to water ratio? **Yes/No**

为何更改 **或** 保留粉水比？

Why did you change the coffee to water ratio, **or** why did you leave the coffee to water ratio unchanged?

解释合理可得 1 分

1 point for suitable explanation

咖啡粉量及用水量 **Ground Coffee & Water Doses**	您是否更改咖啡粉量和/或用水量？**是 / 否** Did you change the ground coffee dose and/or water dose? **Yes/No**

为何改变咖啡粉量和/或用水量？**或是** 为何不改变咖啡粉量和/或用水量？

Why did you change the ground coffee and/or water dose, **or** why did you leave the ground coffee dose and/or water dose unchanged?

解释合理可得 1 分

1 point for suitable explanation

其它参数/变量 **Other Control Measures**	您是否更改任何其它参数/变量？**是 / 否** Did you change any other control measures? **Yes/No**

为何保留所有参数/变量不变？ **或是** 更改了哪些参数/变量及其原因？

Why did you leave all other control measures unchanged, **or** which control measures did you change, and why did you change them?

解释合理可得 1 分

1 point for suitable explanation

总分 = 6 Total points available = 6	总得分 Total points awarded	

第 1 大题 C 部分 咖啡浓度比较
PART 1.C SOLUBLES CONCENTRATION COMPARISON

考试时间：60 分钟 Time available: 60 minutes

- 学员将冲煮“符合”下列标准的目标咖啡。
  Learners will create brews that “hit” the targets listed below.
- 每名学员必须解释如何及为何调整参考用咖啡中的使用参数，以达到咖啡 C1 和 C2 的要求。
  Each learner must provide explanations for how and why they adjusted the parameters used in the reference brew to achieve Brews C1 and C2.

维持一致的咖啡萃取率：20%（+/- 1.00%）

Uniform Solubles Yield: 20% (+/- 1.00%)

- 咖啡 C1 - 1.10% 咖啡浓度（+/- 0.05%）
  Brew C1 - 1.10% Solubles Concentration (+/- 0.05%)

- 咖啡 C2 - 1.60% 咖啡浓度（+/- 0.05%）
  Brew C2 - 1.60% Solubles Concentration (+/- 0.05%)

咖啡 Brew	研磨咖啡粉量 Ground Coffee Dose (g/oz)	用水量 Water Dose (g/oz)	冲煮时间 Brew Time	浓度 Solubles Concentration %	萃取率 Solubles Yield %	成功冲煮咖啡所需的时间：____:____ Time taken to prepare successful brew: ____:____	分数 Points
C 1			:			合格咖啡可得 2 分 2 points for successful brew	

描述与参考用咖啡相比的感官差异： Describe sensory differences compared with reference brew:	
描述得宜可得 1 分 1 point for suitable description	

粉水比 **Coffee to Water Ratio**	您是否更改粉水比？**是 / 否** Did you change the coffee to water ratio? **Yes/No**	
为何更改 **或** 保留粉水比？ Why did you change the coffee to water ratio, **or** why did you leave the coffee to water ratio unchanged?		
解释合理可得 1 分 1 point for suitable explanation		

咖啡粉量及用水量 **Ground Coffee & Water Doses**	您是否更改咖啡粉量和/或用水量？**是 / 否** Did you change the ground coffee dose and/or water dose? **Yes/No**	
为何改变咖啡粉量和/或用水量？**或是** 为何不改变咖啡粉量和/或用水量？ Why did you change the ground coffee and/or water dose, **or** why did you leave the ground coffee dose and/or water dose unchanged?		
解释合理可得 1 分 1 point for suitable explanation		

其它参数/变量 **Other Control Measures**	您是否更改任何其它参数/变量？**是 / 否** Did you change any other control measures? **Yes/No**	
为何保留所有参数/变量不变？ **或是** 更改了哪些参数/变量及其原因？ Why did you leave all other control measures unchanged, **or** which control measures did you change, and why did you change them?		
解释合理可得 1 分 1 point for suitable explanation		
总分 = 6 Total points available = 6	总得分 Total points awarded	

咖啡 **Brew**	研磨咖啡粉量 **Ground Coffee Dose (g/oz)**	用水量 **Water Dose (g/oz)**	冲煮时间 **Brew Time**	浓度 **Solubles Concentration %**	萃取率 **Solubles Yield %**	**成功冲煮咖啡所需的时间：____:____** **Time taken to prepare successful brew: ____:____**	分数 **Points**
C 2			:			合格咖啡可得 2 分 2 points for successful brew	

描述与参考用咖啡相比的感官差异： Describe sensory differences compared with reference brew:	
描述得宜可得 1 分 1 point for suitable description	

粉水比 **Coffee to Water Ratio**	您是否更改粉水比？**是 / 否** Did you change the coffee to water ratio? **Yes/No**	
为何更改 **或** 保留粉水比？ Why did you change the coffee to water ratio, **or** why did you leave the coffee to water ratio unchanged?		
	解释合理可得 1 分 1 point for suitable explanation	

咖啡粉量及用水量 **Ground Coffee & Water Doses**	您是否更改咖啡粉量和/或用水量？**是 / 否** Did you change the ground coffee dose and/or water dose? **Yes/No**	
为何改变咖啡粉量和/或用水量？**或是** 为何不改变咖啡粉量和/或用水量？ Why did you change the ground coffee and/or water dose, **or** why did you leave the ground coffee dose and/or water dose unchanged?		
	解释合理可得 1 分 1 point for suitable explanation	

其它参数/变量 **Other Control Measures**	您是否更改任何其它参数/变量？**是 / 否** Did you change any other control measures? **Yes/No**	
为何保留所有参数/变量不变？ **或是** 更改了哪些参数/变量及其原因？ Why did you leave all other control measures unchanged, **or** which control measures did you change, and why did you change them?		
	解释合理可得 1 分 1 point for suitable explanation	

总分 = 6 Total points available = 6	总得分 Total points awarded	

第 2 大题：冲煮用水分析 Section 2: Water analysis

总分 10 分|总时间 10 分钟 10 total points possible | 10 total minutes

- 本大题所采用的水，与第 1 大题冲煮用水以及水质实践活动（3.03.07.08）三种用水相同。
  Water used for brewing in section 1, and three of the same waters used in the water activity (3.03.07.08) are used in this section.
- 要求学员分析第 1 大题冲煮用水的样本，并陈述其是否符合 SCA 建议。
  The learner analyses a sample of the water used for brewing in section 1 and indicates whether it meets SCA recommendations.
- 接着向学员提供另外 3 份水样本，每份均标示一组三位数代码。
  The learner is then offered 3 additional water samples, each identified with a 3digit code.
- 学员应从这 3 份水样本中择一进行分析，并陈述该样本经测量后是否符合 SCA 建议。本大题可于课程任何阶段完成，由 AST 酌情决定。
  The learner chooses and analyses 1 of these 3 additional samples and indicates whether the measured sample meets SCA recommendations. At the discretion of the AST, this section may be completed at any stage of the course.

冲煮用水分析 Water Analysis						
样本 Sample	**碱度 (每个样本 1 分) Alkalinity (1 point available per sample)**	**总硬度 (每个样本 1 分) Total Hardness (1 point available per sample)**	**pH 值 (每个样本 1 分) (1 point available per sample)**	**电导率 (每个样本 1 分) Electrical Conductivity (1 point available per sample)**	**是否符合 SCA 建议 (每个样本 1 分) Meets SCA Recommendations (1 point available per sample)**	**得分 Points Awarded**
第 1 大题 冲煮用水 Section 1 Water	ppm $CaCO_3$	ppm $CaCO_3$		µS/cm	是/否 Yes/No	
分数 Points						
#_______	ppm $CaCO_3$	ppm $CaCO_3$		µS/cm	是/否 Yes/No	
分数 Points						
				总分 = 10 Total points available = 10	总得分 Total points awarded	

应按以下要求制作/购买水样本：Water samples should be prepared/purchased as follows:				
样本 Sample	**碱度 Alkalinity**	**总硬度 Total Hardness**	**pH 值 pH**	**电导率 Electrical Conductivity**
774	10 至 20 ppm $CaCO_3$ 10 to 20 ppm $CaCO_3$	20 至 30 ppm $CaCO_3$ 20 to 30 ppm $CaCO_3$	6.5 至 8.0 6.5 to 8.0	10 至 1000 µ/cm 10 to 1000µ/cm
913	140 至 200 ppm $CaCO_3$ 140 to 200 ppm $CaCO_3$	>200 ppm $CaCO_3$	6.5 至 8.0 6.5 to 8.0	10 至 1000 µ/cm 10 to 1000µ/cm
642	10 至 35 ppm $CaCO_3$ 10 to 35 ppm $CaCO_3$	>200 ppm $CaCO_3$	6.5 至 8.0 6.5 to 8.0	10 至 1000 µ/cm 10 to 1000µ/cm

第 3 大题：提升冲煮咖啡质量（A、B 和 C 部分）

Section 3: Improve brew quality (parts A, B, and C)

总分 17 分|总时间 45 分钟 17 total points possible | 45 total minutes

- 当学员进入第 3 大题时， AST 必须展示在第 1 大题以及第 2 大题中使用的所有冲煮用水的具体指标，以供学员考虑选择。这些用水都可用于第 3 大题冲煮咖啡。
  When the learner begins section 3, the AST must reveal the water specifications for the water used in section 1, and all of the waters available in Section 2 for consideration. All of these waters will be available for brewing in section 3.
- 咖啡 3.2 的咖啡浓度或萃取率无任何限制。
  There are no restrictions on the solubles concentration or solubles yield for Brew 3.2.

第 3 大题 A 部分 分析和建议

Part 3.A Analysis & proposal

- A **部分**将要求学员回顾自己对第 1 大题咖啡的感官认知，并指出可对第 1 大题参考用咖啡进行的一至多项潜在改进。
  Part A requires the learner to recall their sensory observations from the brews in section 1 and identify one or more potential improvements that could be made to the reference brew from section 1.
- 学员还将自述可在冲煮参数和用水选择等方面做出的更改，以实现前述的改进事项。
  The learner also proposes possible changes that could be made to brewing parameters and water selection to deliver the proposed improvements.
- 接着再由 AST 评估该建议是否合理。
  The AST then assesses whether the proposal is reasonable or not.
- **注意**：如果学员和 AST 一致认为参考用咖啡无需改进，学员可选择提出一种符合标准的冲煮方式，但应凸显与参考用咖啡的质量有哪些不同。
  Note: If the learner and AST agree that the reference brew cannot be improved upon, the learner may choose to propose a brew which could be considered good, but highlight different qualities of the coffee from those presented in the reference brew.

<table>
<tr><th colspan="2">分析和建议
Analysis & Proposal</th><th>分数
Points</th></tr>
<tr><td>根据您对第 1 大题所有咖啡的分析，参考用咖啡哪些方面尚有改进空间？
Based on your analysis of all brews prepared in section 1, which aspects of the reference brew can potentially be improved?</td><td rowspan="4">合理分析和建议
（共 3 分）
Reasonable Analysis & Proposal
(3 points available)</td><td rowspan="4"></td></tr>
<tr><td>示例：我想萃取更多的甜味，增加乾净度，
Example: I'd like to pull more sweetness and get a little more clarity,</td></tr>
<tr><td>因此我会减少研磨咖啡粉量，从而降低浓度并提高萃取率，
so I'll reduce the ground coffee dose to reduce strength and increase the extraction,</td></tr>
<tr><td>也研磨得更细一点。
and also grind a little finer.</td></tr>
<tr><td></td><td>是/否
Yes/No</td><td></td></tr>
</table>

第 3 大题 B 部分 咖啡质量预测

Part 3.B Brew prediction

- **B 部分**将要求学员使用【冲煮参数和参数修正工作表】，根据第 3 大题 A 部份的建议制定冲煮计划，并预测与参考用咖啡相比，咖啡质量有哪些方面将得到改进。计划必须包括用水选择。
  Part B requires the learner to use a Brewing Professional Recipe Modification Worksheet and create a brew plan based on the3.A proposal and predict which aspects of the quality will be improved when compared with the reference brew. The plan must include a water selection.
- 学员预测改进后的计划将如何影响冲煮咖啡的感官质量，以及咖啡萃取率和浓度。
  The learner predicts how the improved plan will affect the sensory qualities, and the solubles yield and solubles concentration of the brew.
- 接着由 AST 评估预测并判定预测是否合理。
  The AST then assesses the predictions and determines if the predictions are reasonable.

<table>
<tr><th colspan="4">咖啡质量预测 Brew Prediction</th><th>分数
Points</th></tr>
<tr><td>用水选择（圈选一项）
Water Choice (Circle one)</td><td colspan="3">水#774 水#913 水#642 第 1 大题中使用的水
Water #774 Water #913 Water #642 Water from Section 1</td><td></td></tr>
<tr><td colspan="3">您的冲煮计划和/或用水选择将如何影响咖啡 3.2 的酸度？
How will your brew plan and/or choice of water affect the acidity of brew 3.2?</td><td>合理预测（1 分）
Reasonable Prediction
(1 point available)</td><td></td></tr>
<tr><td colspan="3">示例：水#774 的碱度较低，会导致咖啡的酸度较高
Example: The lower alkalinity of water #774 will produce brighter acidity</td><td>是 / 否
Yes/No</td><td></td></tr>
<tr><td colspan="3">您的冲煮计划和/或用水选择将如何影响咖啡 3.2 的甜度？
How will your brew plan and/or choice of water affect the sweetness of brew 3.2?</td><td>合理预测（1 分）
Reasonable Prediction
(1 point available)</td><td></td></tr>
<tr><td colspan="3">示例：磨得越细，甜味越浓
Example: The finer grind setting will produce more sweetness</td><td>是 / 否
Yes/No</td><td></td></tr>
<tr><td colspan="3">您的冲煮计划和/或用水选择将如何影响咖啡 3.2 的苦味？
How will your brew plan and/or choice of water affect the bitterness of brew 3.2?</td><td>合理预测（1 分）
Reasonable Prediction
(1 point available)</td><td></td></tr>
<tr><td colspan="3">示例：磨得越细，苦味越浓
Example: The coarser grind setting will produce less bitterness</td><td>是 / 否
Yes/No</td><td></td></tr>
<tr><td colspan="3">您的冲煮计划和/或用水选择将如何影响咖啡 3.2 的风味？
How will your brew plan and/or choice of water affect the flavor of brew 3.2?</td><td>合理预测（1 分）
Reasonable Prediction
(1 point available)</td><td></td></tr>
<tr><td colspan="3">示例：水#913 的 TH 值越高，风味更浓厚，而粉水比越低，风味越淡
Example: The higher TH of water #913 will produce more complex flavour, and the lower coffee to water ratio will make the flavour more transparent</td><td>是 / 否
Yes/No</td><td></td></tr>
<tr><td colspan="3">您的冲煮计划和/或用水选择将如何影响咖啡 3.2 的醇厚度？
How will your brew plan and/or choice of water affect the body of brew 3.2?</td><td>合理预测（1 分）
Reasonable Prediction
(1 point available)</td><td></td></tr>
<tr><td colspan="3">示例：粉水比越低，醇厚度就越低。
Example: The lower coffee to water ratio will produce a lighter bodied bre</td><td>是 / 否
Yes/No</td><td></td></tr>
<tr><td colspan="3">您的冲煮计划和/或用水选择将如何影响咖啡 3.2 的浓度？（圈选一项）
How will your brew plan and/or choice of water affect the solubles concentration of brew 3.2? (Circle one)</td><td rowspan="2">合理预测（1 分）
Reasonable Prediction
(1 point available)
是 / 否
Yes/No</td><td rowspan="2"></td></tr>
<tr><td>无影响（差异小于 0.05%）
No effect (less than 0.05% difference)</td><td>浓度更低
Lower Concentration</td><td>浓度更高
Higher Concentration</td></tr>
<tr><td colspan="3">您的冲煮计划和/或用水选择将如何影响咖啡 3.2 的萃取率（圈选一项）
How will your brew plan and/or choice of water affect the solubles yield of brew 3.2?(Circle one)</td><td rowspan="2">合理预测（1 分）
Reasonable Prediction
(1 point available)
是 / 否
Yes/No</td><td rowspan="2"></td></tr>
<tr><td>无影响（差异小于 0.5%）
No effect (less than 0.5% difference)</td><td>萃取率更低
Lower Yield</td><td>萃取率更高
Higher Yield</td></tr>
<tr><td colspan="2">总分 = 7
Total points available = 7</td><td colspan="2">总得分
Total points awarded</td><td></td></tr>
</table>

第 3 大题 C 部分 冲煮咖啡成果 Part 3.C Brew outcome

- **C 部分**要求学员冲煮两杯咖啡： **Part C** requires the learner to prepare two brews:
  - *咖啡 3.1*：使用第 1 大题参考用咖啡的配方和水，进行参考用咖啡冲煮，并满足指定的咖啡浓度和萃取率：浓度为 1.30%（+/-0.05%），萃取率为 20%（+/-1.00%）。
    Brew 3.1: a reference brew using the reference brew recipe and water as used in section 1, and meeting target solubles concentration and solubles yield specified: a solubles concentration of 1.30% (+/- 0.05%), with a solubles yield of 20% (+/-1.00%).
  - *咖啡 3.2*：根据 B 部分详述的冲煮计划和用水选择，进行改进修正冲煮。咖啡 3.2 沒有设定咖啡浓度或萃取率目标。
    Brew 3.2: an improved brew based on the brew plan and water selection detailed in Part B. There are no solubles concentration or solubles yield targets for brew 3.2.

  注意：如果需要详细执行冲泡计划，学员可充煮多杯咖啡，但不得更改冲煮计划细节或预测。
  Note: The learner may prepare more than one brew if needed to execute the brew plans as detailed but may not change the details of the brew plan or their predictions.
- 要求学员于测量和评估咖啡 3.1 和 3.2 后，完成技术分析和描述咖啡 3.1 和 3.2 的感官差异。
  The learner measures and assess brews 3.1 and 3.2 and completes technical analysis and describes sensory differences between brew 3.1 and brew 3.2.
- 接着由 AST 评估咖啡 3.1 和 3.2，判定学员的描述是否合理，并评价学员在 B 部分预测结果方面的实现程度。
  The AST then assesses brews 3.1 and 3.2, determines if the learner's descriptions are reasonable, and evaluates how well the learner was able to achieve the results predicted in Part B.

冲煮咖啡成果 Brew Outcome						**分数 Points**
咖啡 3.1 Brew 3.1	咖啡浓度 Solubles Concentration	%	咖啡萃取率 Solubles Yield	%	测量正确（1 分） 是 / 否 Measured correctly Yes/No (1 point available)	
咖啡 3.2 Brew 3.2	咖啡浓度 Solubles Concentration	%	咖啡萃取率 Solubles Yield	%	测量正确（1 分） 是 / 否 Measured correctly Yes/No (1 point available)	
描述咖啡 3.1 和 3.2 的酸度差异 Describe the difference in **acidity** between Brew 3.1 and Brew 3.2 *示例：咖啡 3.2 的酸度较高* *Example: The acidity in Brew 3.2 is brighter*					合理描述（1 分） Reasonable description (1 point available)	
描述咖啡 3.1 和 3.2 的甜度差异 Describe the difference in **sweetness** between Brew 3.1 and Brew 3.2 *示例：冲煮咖啡 3.2 的甜味较浓* *Example: The sweetness in Brew 3.2 is more intense*					合理描述（1 分） Reasonable description (1 point available)	
描述咖啡 3.1 和 3.2 的苦味差异 Describe the difference in **bitterness** between Brew 3.1 and Brew 3.2 *示例：冲煮咖啡 3.2 的苦味较淡* *Example: The bitterness in Brew 3.2 is milder*					合理描述（1 分） Reasonable description (1 point available)	
描述咖啡 3.1 和 3.2 的风味差异 Describe the difference in **flavour** between Brew 3.1 and Brew 3.2 *示例：冲煮咖啡 3.2 的风味较丰富* *Example: The flavour of Brew 3.2 is more complex*					合合理描述（1 分） Reasonable description (1 point available)	
描述咖啡 3.1 和 3.2 的醇厚度差异 Describe the difference in **body** between Brew 3.1 and Brew 3.2 *示例：咖啡 3.2 的醇厚度较低，类似茶的醇厚度* *Example: The body of Brew 3.2 lighter and more tea like*					合理描述（1 分） Reasonable description (1 point available)	
总分 = 7 Total points available = 7					总得分 Total points awarded	

考试得分小计 Exam Points Summary				
第 1 大题 – 分数 Section 1 - Points	总分 Points Available	得分 Points Awarded	已提交冲煮计划 Brew Plan submitted	
参考用咖啡 - A Reference Brew - A	7		是 Yes	否 No
咖啡 B1 Brew B1	6		是 Yes	否 No
咖啡 B2 Brew B2	6		是 Yes	否 No
咖啡 C1 Brew C1	6		是 Yes	否 No
咖啡 C2 Brew C2	6		是 Yes	否 No
第 1 大题 - 总分 本大题及格分数：25 分 Section 1 - Total points Minimum 25 points required to pass section.	31		注意：必须提交所有冲煮计划始予计分。 Note: All brew plans must be submitted for points to be awarded.	

第 2 大题 – 分数 Section 2 - Points	总分 Points Available	得分 Points Awarded
第 2 大题 - 总分 本大题及格分数：8 分 Section 2 - Total points Minimum 8 points required to pass section.	10	

第 3 大题 - 分数 Section 3 - Points	总分 Points Available	得分 Points Awarded	已提交冲煮计划 Brew Plan submitted	
A 部分 Part A	3		是 Yes	否 No
B 部分 Part B	7			
C 部分 Part C	7			
第 3 大题 - 总分 本大题及格分数：12 分 Section 3 - Total point Minimum 12 points required to pass section.	17		注意：必须提交所有冲煮计划始予计分。 Note: All brew plans must be submitted for points to be awarded.	

	圈选一项 Circle One	
第 1 大题 –及格 Pass Section 1	否 No	是 Yes
第 2 大题 - 及格 Pass Section 2	否 No	是 Yes
第 3 大题 –及格 Pass Section 3	否 No	是 Yes
实操考试 - 及格 （必須通過所有測驗） Pass Practical Exam (All sections must be passed)	**否 No**	**是 Yes**